

1 The Beta(α, β) Distribution

2 In the previous Exercise, the random variable T has the Beta($\alpha = 4, \beta = 1$)
3 Distribution.

4 The random variable X has the **Beta(α, β) Distribution** if it has density

$$5 \quad f_X(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$\alpha, \beta > 0$

6 This is a useful model for proportions or probabilities.

7 Note that $E(X) = \alpha/(\alpha + \beta)$.

8 Also note that if $\alpha = \beta = 1$, then the distribution is Uniform(0,1). Further,
9 if $\alpha = \beta$ then the distribution is symmetric about 0.5.

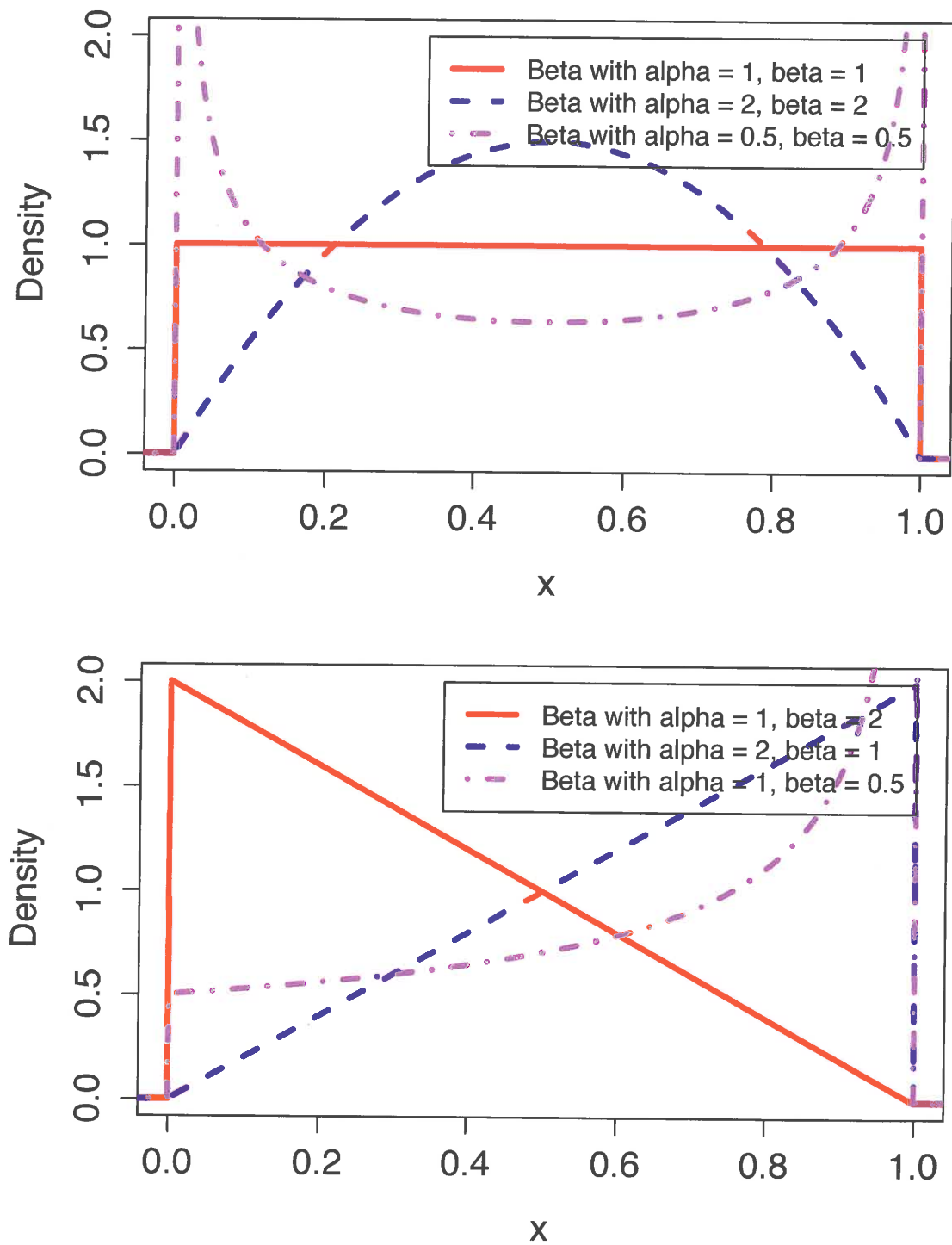


Figure 1: Beta Distribution pdfs.

Curve Fitting of the Corporate Recovery Rates: The Comparison of Beta Distribution Estimation and Kernel Density Estimation

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Abstract

Recovery rate is essential to the estimation of the portfolio's loss and economic capital. Neglecting the randomness of the distribution of recovery rate may underestimate the risk. The study introduces two kinds of models of distribution, Beta distribution estimation and kernel density distribution estimation, to simulate the distribution of recovery rates of corporate loans and bonds. As is known, models based on Beta distribution are common in daily usage, such as CreditMetrics by J.P. Morgan, Portfolio Manager by KMV and Losscalc by Moody's. However, it has a fatal defect that it can't fit the bimodal or multimodal distributions such as recovery rates of corporate loans and bonds as Moody's new data show. In order to overcome this flaw, the kernel density estimation is introduced and we compare the simulation results by histogram, Beta distribution estimation and kernel density estimation to reach the conclusion that the Gaussian kernel density distribution really better imitates the distribution of the bimodal or multimodal data samples of corporate loans and bonds. Finally, a Chi-square test of the Gaussian kernel density estimation proves that it can fit the curve of recovery rates of loans and bonds. So using the kernel density distribution to precisely delineate the bimodal recovery rates of bonds is optimal in credit risk management.

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Introduction

Credit risk is the distribution of financial losses caused by unexpected changes in compliance of financial agreements. The recovery rate is an important measure of how much we can retrieve from bad debts. So it is crucial to figure out what kind of distribution recovery rates of the sample comply with. An important key in building the credit risk model is the recovery rate in default or loss given default (LGD) function, expressed as a ratio (dollar recovery/amount invested) [1].

Recently, research on recovery rate has mainly focused on factors that impact recovery rate, correlation between recovery rate and default rate, its distribution, etc. [2]. Compared with default rate, the influencing factors of recovery rate are more complex. The most representative model is Losscalc Model of Moody's. The correlation between recovery rate and default rate turns out to be positive. Hu & Perraudin [3], Rosch & Scheule [4] find that it will underestimate the portfolio's loss if the correlation is neglected.

Histogram is common and is widely used to depict the distribution, but it can't be a smooth curve [5]. Beta distribution estimation is widely used for simulating the curve of the recovery rates [6]. Besides, Frye [7] assumes that the recovery rate follows a normal distribution and Pykhtin [8] establishes a log-normal distribution and Andersen & Siddiqui [9] discuss the probit-normal distribution. Düllmann & Trapp [10] utilize a logit-normal

distribution and empirically analyse the recovery rates. After comparing the results with two other extended models, Frye [7] and Pykhtin [8], the log-normal distribution is not found to be as good as the other two distributions.

How to precisely depict the randomness of recovery rate is essential to the estimation of the loss of portfolio and economic capital. In this paper, we mainly focus on distribution of recovery rate. Beta distribution is widely used in many credit risk models such as CreditMetrics by J.P. Morgan, Portfolio Manager by KMV and Losscalc by Moody's. However, sometimes it ceases to be effective because of the fact that the recovery rates curve always has two peaks while Beta distribution estimation can only demonstrate one peak. So in this paper we adopt the Gaussian kernel density estimation, a nonparametric estimation, to solve this problem and justify our hypothesis. Kernel density estimation put forward by Emanuel Parzen (1955) and Murray Rosenblatt (1962), also named as the Parzen-Rosenblatt window method, is a non-parametric way to estimate the probability density function of a random variable. Ruppert and Cline proposed revision of the kernel density estimation method based upon data density function and clustering algorithm. We can build the prediction model of value at risk through the kernel density estimation of a random variable. Moreover, weighted processing of the estimated variation coefficient can help build different risk prediction

Marginal Distributions

If one starts with the joint distribution (pmf or pdf) for (X, Y) , it is possible to derive the distribution for X ignoring Y and the distribution for Y ignoring X .

These are called the **marginal distributions** for X and Y , respectively.

These are simply the f_X and f_Y already described.

In the **discrete case**, you “sum over” the other random variable, i.e.,

$$f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f_{X,Y}(x, y)$$

Exercise: Find f_X and f_Y in the Insurance example above.

$$P(X=100) = P(X=100, Y=0) + P(X=100, Y=100) + P(X=100, Y=200)$$

$$= 0.20 + 0.10 + 0.20 = 0.5$$

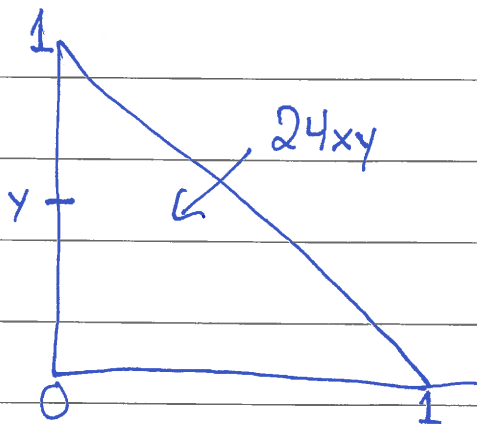
$$P(X=250) = 0.5$$

1 In the **continuous case**, you “integrate out” the other variable:

$$2 \quad f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

3 **Exercise:** Return to our “triangular support” example. Find f_X and f_Y .

$$4 \quad f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y) dx$$

$$5 \quad = \int_0^{1-y} 24xy dx$$


6 (when $0 \leq y \leq 1$)

$$7 \quad = 12y(1-y)^2, \quad 0 \leq y \leq 1$$

8 Likewise, $f_X(x) = 12x(1-x)^2, \quad 0 \leq x \leq 1$

$$9 \quad Y \sim \text{Beta}(\alpha=2, \beta=3)$$

$$10 \quad X \sim \text{Beta}(\alpha=2, \beta=3)$$

- 1 **Exercise:** An electronic system has one each of two different types of com-
 2 ponents in joint operation. Let X and Y denote the random lengths of life
 3 of the components of type I and II, respectively. The joint function is:

$$f_{X,Y}(x,y) = \begin{cases} 8xe^{-2(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

5 Find the marginal pdf $f_X(x)$ and $f_Y(y)$.

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$= \int_0^{\infty} 8xe^{-2(x+y)} dy, \quad x > 0$$

$$= 8xe^{-2x} \int_0^{\infty} e^{-2y} dy, \quad x > 0$$

$$= 4xe^{-2x}, \quad x > 0$$

$$X \sim \text{Gamma}(\alpha=2, \beta=2)$$

$$\text{Also, } f_Y(y) = 2e^{-2y}, \quad y > 0$$

$$\text{So, } Y \sim \text{exponential}(2)$$

1 In the above example, we can see that

$$2 \quad f_{X,Y}(x,y) = f_X(x)f_Y(y), \text{ for all } x,y$$

3 This implies that X and Y are **independent**.

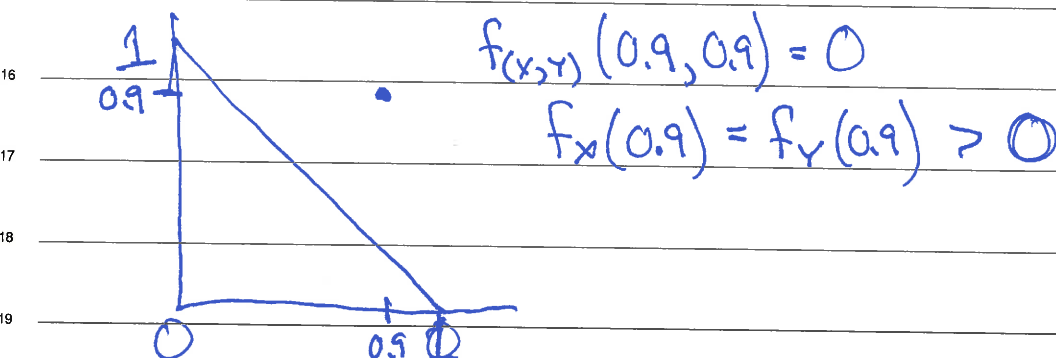
4 The above can be taken as the definition of independent random variables
5 in both the discrete and continuous cases. Further, we say that $X_1, X_2, \dots, X_n$
6 are independent if for all $x_1, x_2, \dots, x_n$,

$$7 \quad f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

8 **Exercise:** Consider a case where (X, Y) have a continuous bivariate distri-
9 bution. Is “the support of the distribution is rectangular” a necessary or
10 a sufficient condition for X and Y being independent?

11 Definitely not sufficient, consider bivariate normal

12
13 It is necessary. I.e., cannot find an example
14 where (X, Y) has non-rectangular support
15 and X, Y are independent.



1 Rule for the Lazy Statistician

2 For a general function $g(x, y)$, one can find expected values using

$$3 \quad E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

4 **Exercise:** Try the case $g(x, y) = x$ to verify that the result is $E(X)$.

$$5 \quad E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

$$7 \quad = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy$$

$$10 \quad = \int_{-\infty}^{\infty} x \underbrace{\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy}_{f_X(x)} dx$$

$$13 \quad = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$16 \quad = E(X)$$

Exercise: Assume that X and Y are independent.

Show that $E(h_1(X)h_2(Y)) = E(h_1(X))E(h_2(Y))$.

$$E(h_1(X)h_2(Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x)h_2(y) f_{X,Y}(x,y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x)h_2(y) f_X(x)f_Y(y) dx dy$$

since X, Y independent

$$= \left[\int_{-\infty}^{\infty} h_1(x) f_X(x) dx \right] \left[\int_{-\infty}^{\infty} h_2(y) f_Y(y) dy \right]$$

$$= E(h_1(X)) E(h_2(Y))$$

Comment: $E(XY) = E(X)E(Y)$ means

X, Y are uncorrelated. The conclusion above is much stronger.

Exercise: Assume that U_1 and U_2 are independent uniform(0, 1) random variables. Let $U_{(1)}$ denote the minimum of U_1 and U_2 , and let $U_{(2)}$ denote the larger of those two. Finally, let order statistics $X_{(1)}, X_{(2)}, \dots$

$$L_1 = U_{(1)}, \quad L_2 = U_{(2)} - U_{(1)}, \quad \text{and} \quad L_3 = 1 - U_{(2)}.$$

Note that this is the "stick-broken-into-three-pieces" example, with L_i denoting the lengths of the three pieces ordered from left to right.

1. What is the joint distribution of (U_1, U_2) ?
2. What is the joint distribution of $(U_{(1)}, U_{(2)})$?
3. What is the joint distribution of (L_1, L_2) ?
4. Show that L_1 , L_2 , and L_3 each have the Beta(1, 2) distribution. (Do not take it as "obvious" that all three have the same distribution.)

Comment: It is not difficult to see that if k uniform cuts are made, the length of each piece has the Beta(1, k) distribution.

① Joint dist. for (U_1, U_2)

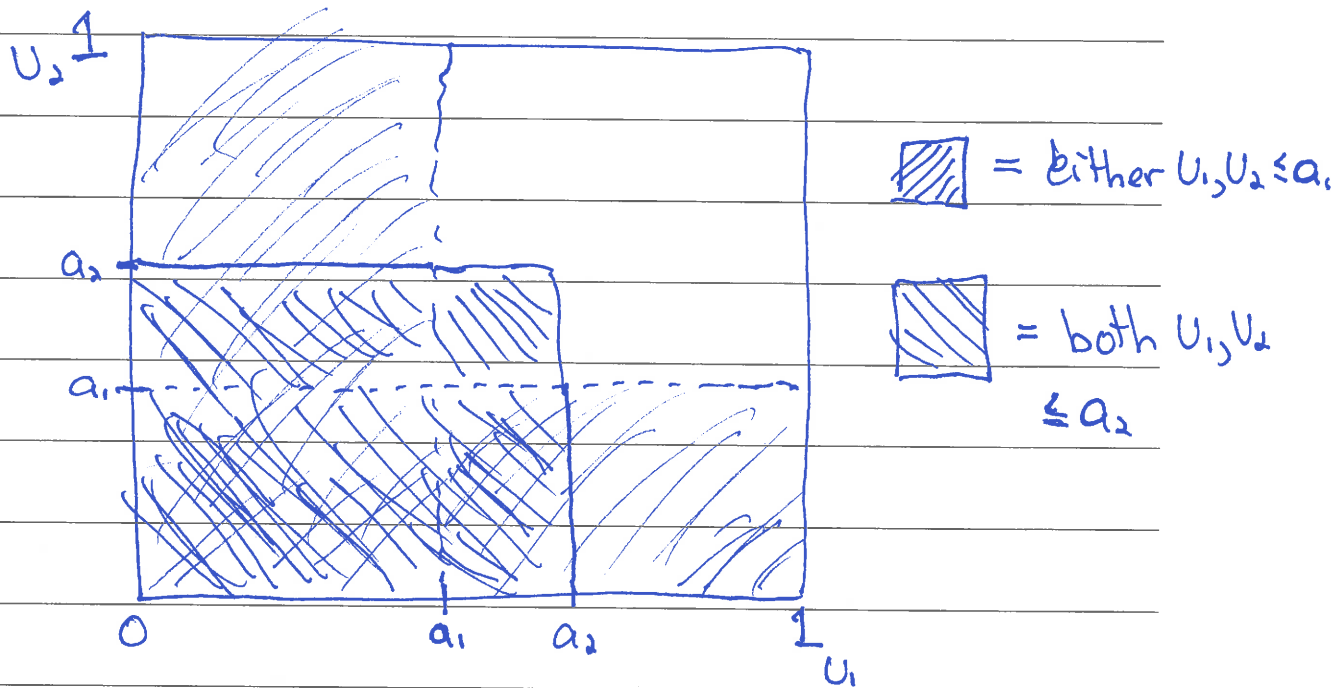
$$f_{(U_1, U_2)}(u_1, u_2) = f_{U_1}(u_1) f_{U_2}(u_2) \quad \text{since independent}$$

$$= \begin{cases} 1, & 0 \leq u_1 \leq 1, 0 \leq u_2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

② What is joint dist. for $(U_{(1)}, U_{(2)})$?

Find the CDF for $(U_{(1)}, U_{(2)})$

$$F_{(U_{(1)}, U_{(2)})}(a_1, a_2) = P(U_{(1)} \leq a_1, U_{(2)} \leq a_2)$$



Need at least one of U_1, U_2 to be less than a_1 , but need both U_1 and U_2 to be less than a_2

If $0 \leq a_1 \leq a_2 \leq 1$, then $= a_1^2 + 2(a_2 - a_1)a_1$
 $= 2a_1a_2 - a_1^2$

If $0 \leq a_2 \leq a_1 \leq 1$, then $= a_2^2$

$P(U_{(1)} \leq 0.9, U_{(2)} \leq 0.1) = P(U_1, U_2 \leq 0.1)$
 $= (0.1)^2$

If $a_1 < 0$ or $a_2 < 0$, then $= 0$

If $a_1 > 1$ and $a_2 > 1$, then $= 1$

If $a_1 > 1$ and $0 \leq a_2 \leq 1$, then $= a_2^2$

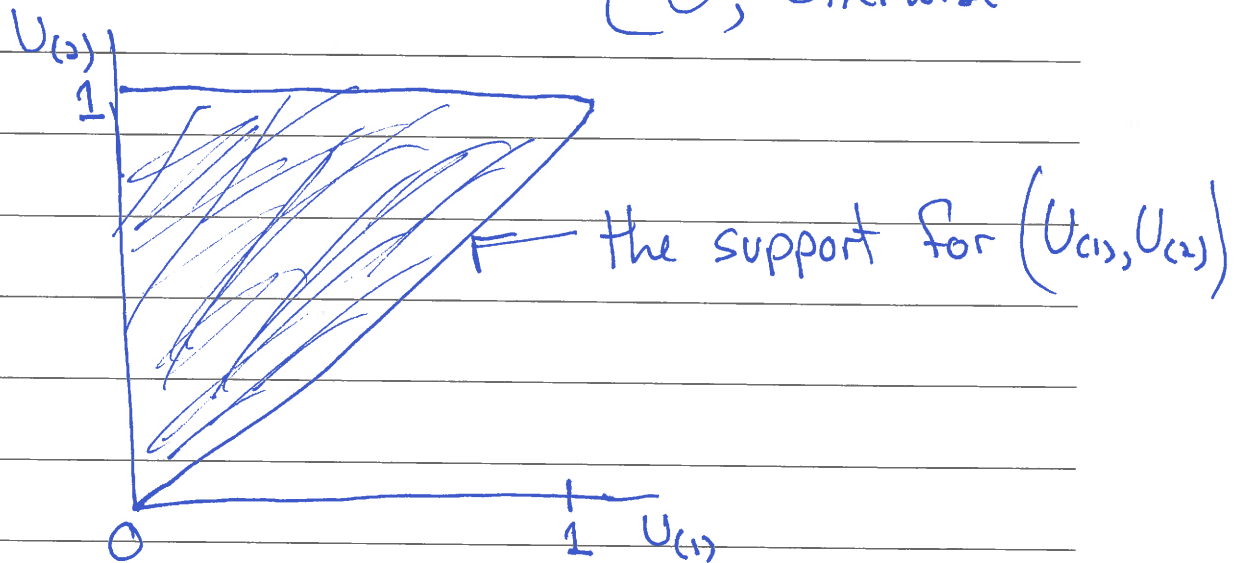
If $0 \leq a_1 \leq 1$ and $a_2 > 1$, then $= P(\text{at least 1 of } U_1, U_2 \leq a_1)$
 $= 1 - (1 - a_1)^2$
 $= 2a_1 - a_1^2$

This completely specifies the CDF for $(U_{(1)}, U_{(2)})$

$$P(0.2 \leq U_{(1)} \leq 0.3, U_{(2)} \leq 0.1) = 0$$

Take the derivative to get the pdf

$$\frac{\partial F_{U_{(1)}, U_{(2)}}(a_1, a_2)}{\partial a_1 \partial a_2} = \begin{cases} 2, & 0 \leq a_1 \leq a_2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



Note that $L_1 = U_{(1)}$, so

$$f_{L_1}(l_1) = \int_{-\infty}^{\infty} f_{(U_{(1)}, U_{(2)})}(a_1, a_2) da_2$$

The Conditional Distributions

If we know (condition on) $Y = y$, what is the distribution of X ? Such a question is answered by considering the **conditional distribution** of X given Y .

In the **discrete case**,

$$P(X=x, Y=y) \quad P(Y=y)$$

$$f_{X|Y}(x|y) = P(X = x|Y = y) = \begin{cases} f_{X,Y}(x, y) / f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases}$$

and in the **continuous case** we have

$$f_{X|Y}(x|y) = \begin{cases} f_{X,Y}(x, y) / f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases},$$

with $f_{X|Y}$ called the **conditional density of X given Y** .

Comment: The continuous case is notable in that the event being conditioned on ($Y = y$) has probability zero. There is nothing mathematically or conceptually incorrect about conditioning on probability zero events.

Exercise: Derive the conditional distribution of X given Y in our "triangular support" example.

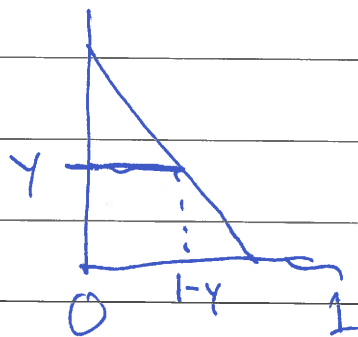
$$f_{X,Y}(x,y) = 24xy, \quad x > 0, y > 0, x+y < 1$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \begin{cases} \frac{24xy}{12y(1-y)^2}, & 0 < y < 1 \\ & 0 < x < 1-y \\ 0, & \text{otherwise} \end{cases}$$

Note that when condition on $Y=y$, $0 < y < 1$,

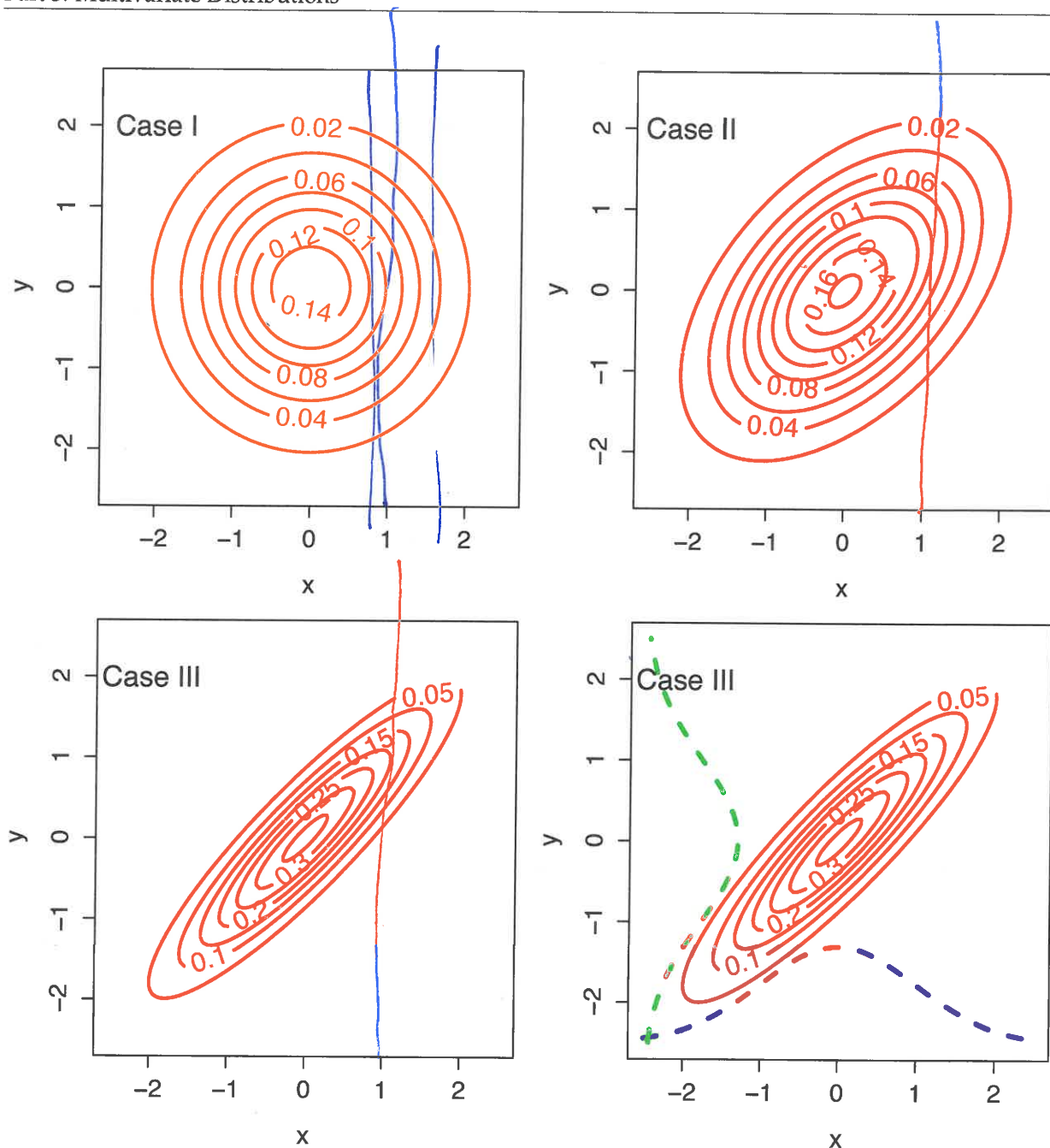
the support for X is 0 to $1-y$



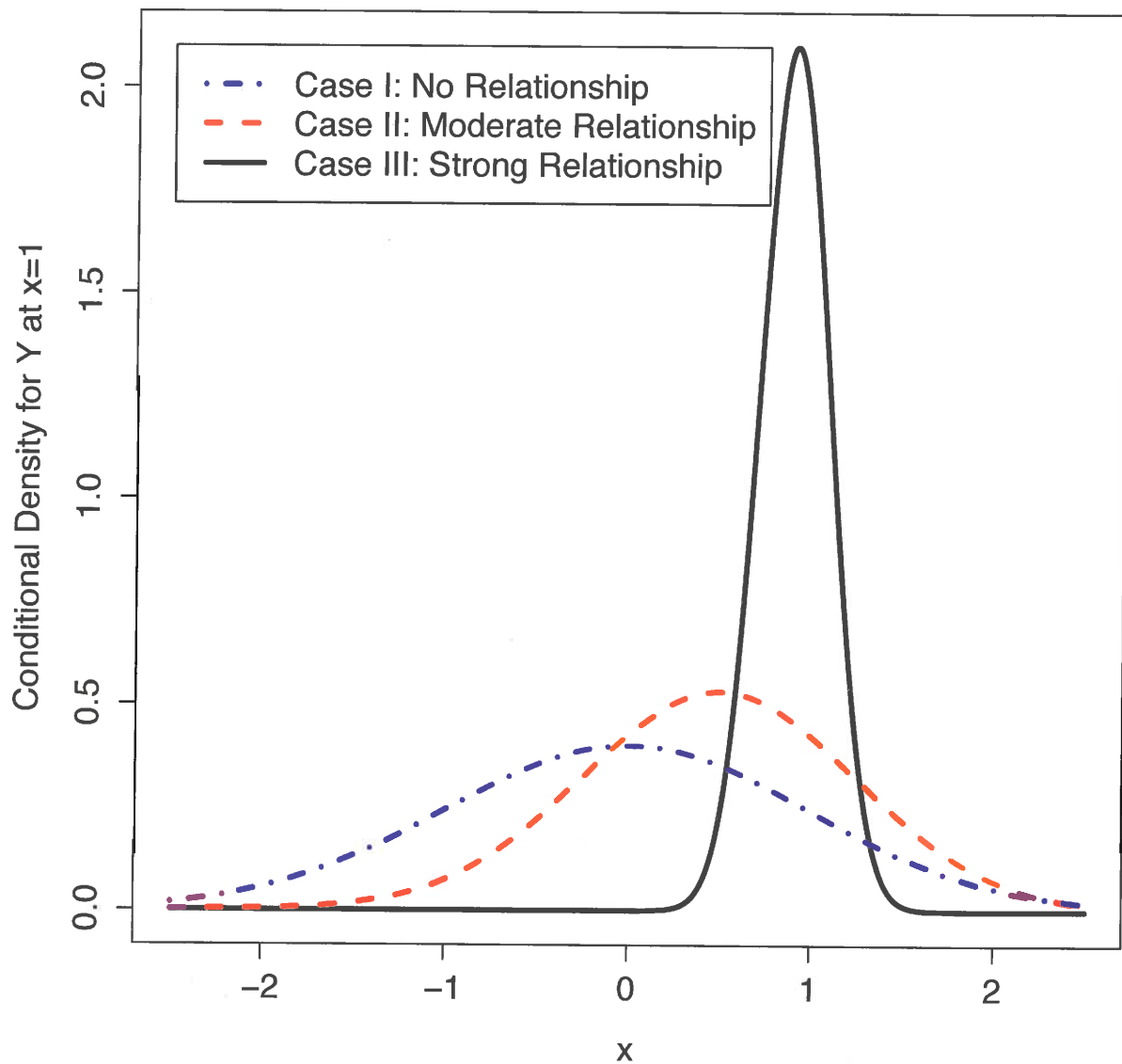
1 **Example:** This example illustrates the distinction between the marginal and
2 conditional distributions. Here, the bivariate pdf is depicted using **contour**
3 **plots**.

4 In **Case I** shown below, X and Y are independent, and every conditional
5 distribution is the same. In **Case II** there is a moderate amount of depen-
6 dence. In **Case III** the dependence is stronger.

7 The fourth plot again shows Case III, but with the marginals imposed. In
8 fact, the marginals are the same in all three Cases.



- 1 The conditional distribution Y for $X = 1$ changes across the three Cases.



2

Covariance

The **covariance between X and Y** , denoted $\text{Cov}(X, Y)$, is defined as

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

This is a measure of the amount of **linear association** between X and Y .

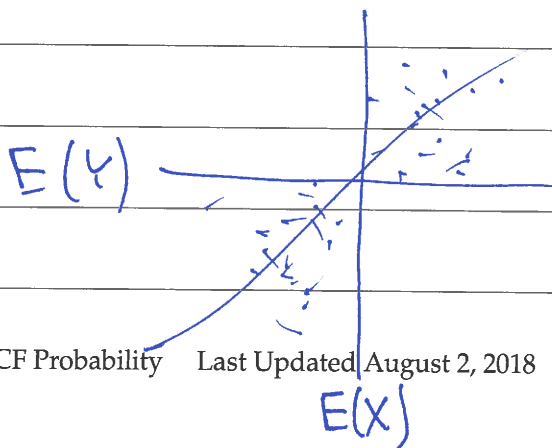
The sign of $\text{Cov}(X, Y)$ tells us the **direction** of that association.

Exercise: Looking at the definition of the covariance, explain under what conditions $\text{Cov}(X, Y) > 0$, $\text{Cov}(X, Y) < 0$, and $\text{Cov}(X, Y) = 0$.

Why would $\text{Cov}(X, Y) > 0$?

If when X exceeds its mean, Y also exceeds its mean, and ~~if~~ when X is less than its mean, Y is also less than its mean

$$(X - E(X))(Y - E(Y)) > 0$$



Can make a similar statement for
 $\text{Cov}(X, Y) < 0$.

When $\text{Cov}(X, Y) = 0$?

If X, Y independent,

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E(X - E(X))E(Y - E(Y))$$

since X, Y independent

$$= 0$$

But, if X, Y are uncorrelated, cannot claim
 X, Y are independent

~~Recall~~ Example: $Z \sim N(0, 1)$, $U = \pm 1$ w.p. $1/2$
 U, Z independent, $Y = UZ$

What is dist of Y ? $Y \sim N(0, 1)$ (use CDF)

$$\begin{aligned} \text{Cov}(Y, Z) &= E(YZ) - E(Y)E(Z) = E(YZ) = E(UZ^2) \\ &= E(U)E(Z^2) = 0, \text{ but } Y, Z \text{ are clearly dependent} \end{aligned}$$

True: When (Y, Z) are bivariate normal,
then $\text{Cov}(Y, Z) = 0 \Rightarrow Y, Z$ independent

But, Y normal, Z normal $\nRightarrow (Y, Z)$
bivariate normal

1 When $\text{Cov}(X, Y) = 0$, then X and Y are said to be **uncorrelated**.

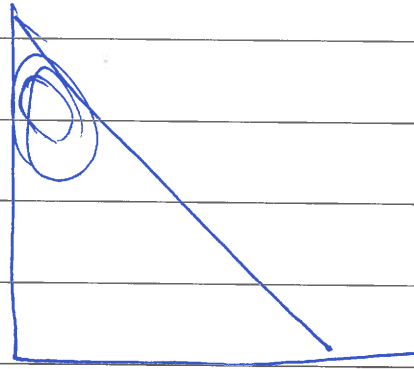
2 Further, the **correlation between X and Y** is defined to be

$$3 \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

4 Properties that you should know:

- 5 1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- 6 2. $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$
- 7 3. $\text{Cov}(X, X) = V(X)$
- 8 4. $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- 9 5. If X and Y are independent, then $\text{Cov}(X, Y) = 0$.
- 10 6. $-1 \leq \text{Corr}(X, Y) \leq 1$

Exercise: Find $\text{Cov}(X, Y)$ for the "triangular support" example.



Recall: $X, Y \sim \text{Beta}(2, 3)$

$$E(X) = E(Y) = \frac{2}{5}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy$$

$$= \int_{y=0}^1 \int_{x=0}^{1-y} xy \cdot 24xy dx dy$$

$$= \frac{8}{60}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{2}{75}$$

Exercise: Suppose that X and Y are random variables with $Y = aX + b$ for constants a and b with $a \neq 0$. What is the correlation between X and Y ?

$$\text{Cov}(X, Y) = \text{Cov}(X, aX + b)$$

$$= a \text{Cov}(X, X) \quad \text{using prop. 4}$$

$$= a V(X)$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

$$V(Y) = V(aX + b) = a^2 V(X)$$

$$= \frac{a V(X)}{\sqrt{a^2 V(X)V(X)}} = \frac{a}{|a|}$$

$$= \text{sgn}(a) = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases}$$